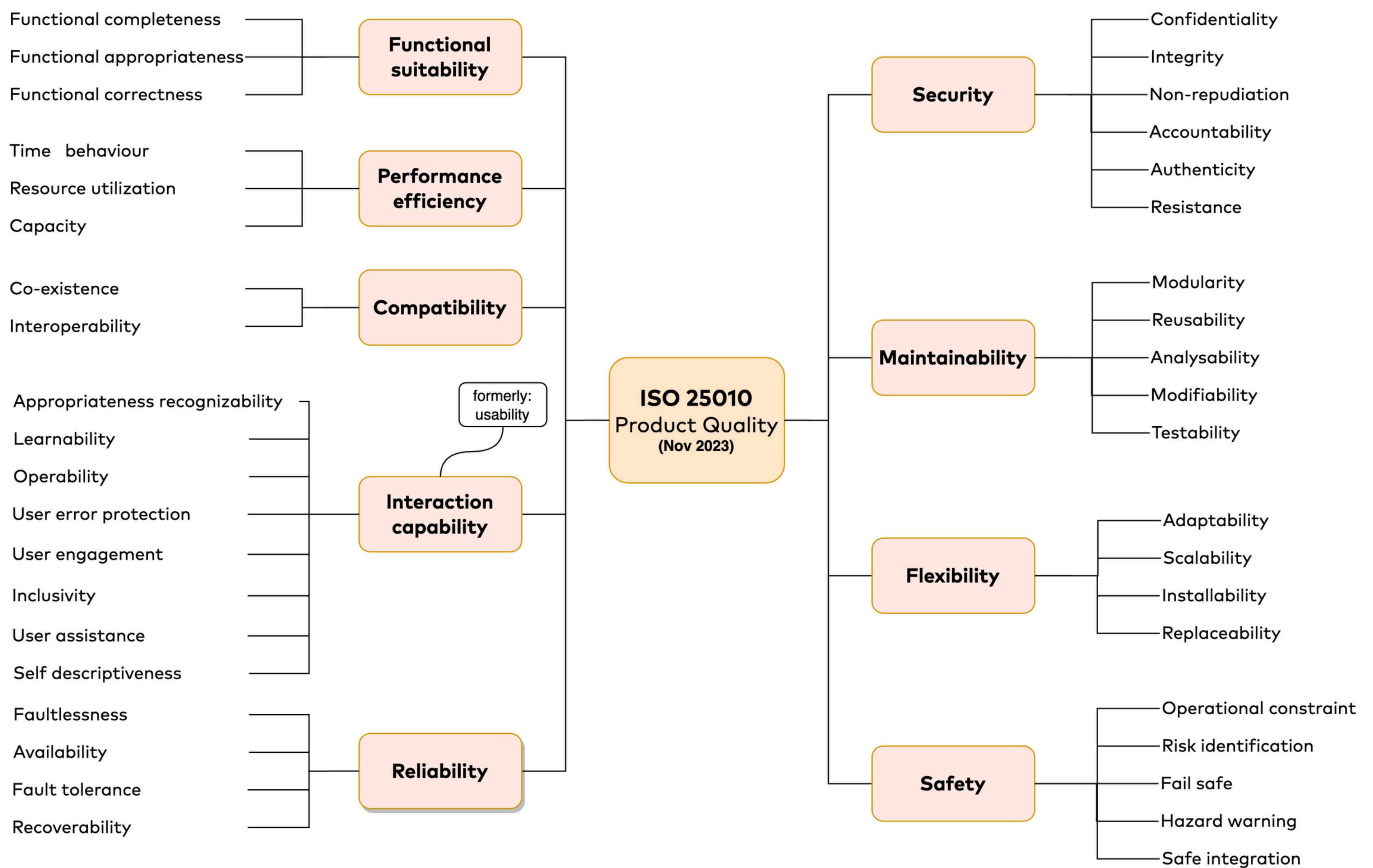


ISO/IEC 25010



ISO/IEC 25010 is a standard that defines a quality model for software product evaluation. It provides a framework to assess the quality of software products and systems based on various characteristics that contribute to their overall quality. This model is part of the ISO/IEC 25000 series, which focuses on software product quality requirements and evaluation.



Functional suitability

— capability of a product to provide functions that meet stated and implied needs of intended users when it is used under specified conditions

Functional completeness

capability of a product to provide a set of functions that covers all the specified tasks and intended users' objectives

Functional correctness

capability of a product to provide accurate results when used by intended users

Functional appropriateness

capability of a product to provide functions that facilitate the accomplishment of specified tasks and objectives

Performance efficiency

```
graph TD; PE[Performance efficiency] --- V[ ]; V --> TB[Time behaviour]; V --> RU[Resource utilization]; V --> C[Capacity]; TB --- TBDef[capability of a product to perform its specified function under specified conditions so that the response time and throughput rates meet the requirements]; RU --- RUDef[capability of a product to use no more than the specified amount of resources to perform its function under specified conditions]; C --- CDef[capability of a product to meet requirements for the maximum limits of a product parameter];
```

capability of a product to perform its functions within specified time and throughput parameters and be efficient in the use of resources under specified conditions

Time behaviour

capability of a product to perform its specified function under specified conditions so that the response time and throughput rates meet the requirements

Resource utilization

capability of a product to use no more than the specified amount of resources to perform its function under specified conditions

Capacity

capability of a product to meet requirements for the maximum limits of a product parameter

Compatibility

```
graph TD; A[Compatibility] --- B[Co-existence]; A --- C[Interoperability];
```

capability of a product to exchange information with other products, and/or to perform its required functions while sharing the same common environment and resources

Co-existence

capability of a product to perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product

Interoperability

capability of a product to exchange information with other products and mutually use the information that has been exchanged

Interaction capability

capability of a product to be interacted with by specified users to exchange information between a user and a system via the user interface to complete the intended task

Appropriateness recognizability

capability of a product to be recognized by users as appropriate for their needs

Learnability

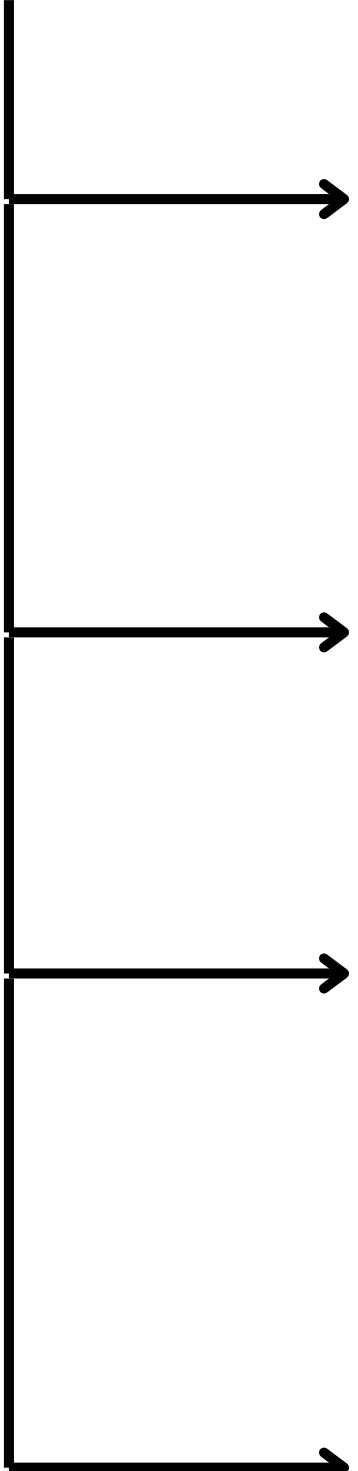
capability of a product to have specified users learn to use specified product functions within a specified amount of time

Operability

capability of a product to have functions and attributes that make it easy to operate and control

User error protection

capability of a product to prevent operation errors



User engagement

capability of a product to present functions and information in an inviting and motivating manner encouraging continued interaction

Inclusivity

capability of a product to be utilised by people of various backgrounds

User assistance

capability of a product to be used by people with the widest range of characteristics and capabilities to achieve specified goals in a specified context of use

Self descriptiveness

capability of a product to present appropriate information, where needed by the user, to make its capabilities and use immediately obvious to the user without excessive interactions with a product or other resources

Reliability

```
graph TD; Reliability[Reliability] --- Line[ ]; Line --> Faultlessness[Faultlessness]; Line --> Availability[Availability]; Line --> FaultTolerance[Fault tolerance]; Line --> Recoverability[Recoverability];
```

capability of a product to perform specified functions under specified conditions for a specified period of time without interruptions and failures

Faultlessness

capability of a product to perform specified functions without fault under normal operation

Availability

capability of a product to be operational and accessible when required for use

Fault tolerance

capability of a product to operate as intended despite the presence of hardware or software faults

Recoverability

capability of a product in the event of an interruption or a failure to recover the data directly affected and re-establish the desired state of the system

Security

```
graph LR; Security[Security] --- Line[ ]; Line --> Confidentiality[Confidentiality]; Line --> Integrity[Integrity]; Line --> NonRepudiation[Non-repudiation];
```

capability of a product to protect information and data so that persons or other products have the degree of data access appropriate to their types and levels of authorization, and to defend against attack patterns by malicious actors

Confidentiality

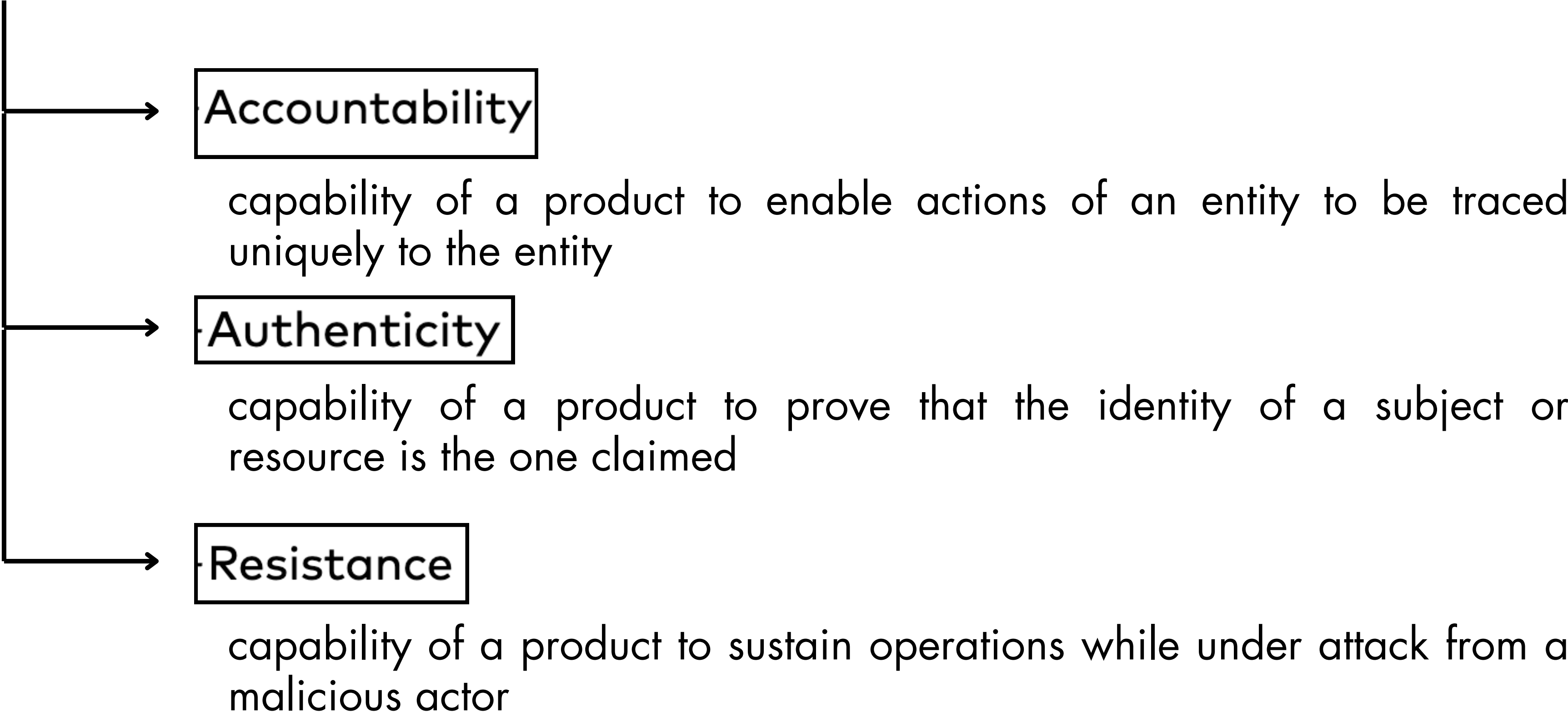
capability of a product to ensure that data are accessible only to those authorized to have access

Integrity

capability of a product to ensure that the state of its system and data are protected from unauthorized modification or deletion either by malicious action or computer error

Non-repudiation

capability of a product to prove that actions or events have taken place, so that the events or actions cannot be repudiated later



```
graph LR; Root[ ] --- A[Accountability]; Root --- B[Authenticity]; Root --- C[Resistance];
```

Accountability

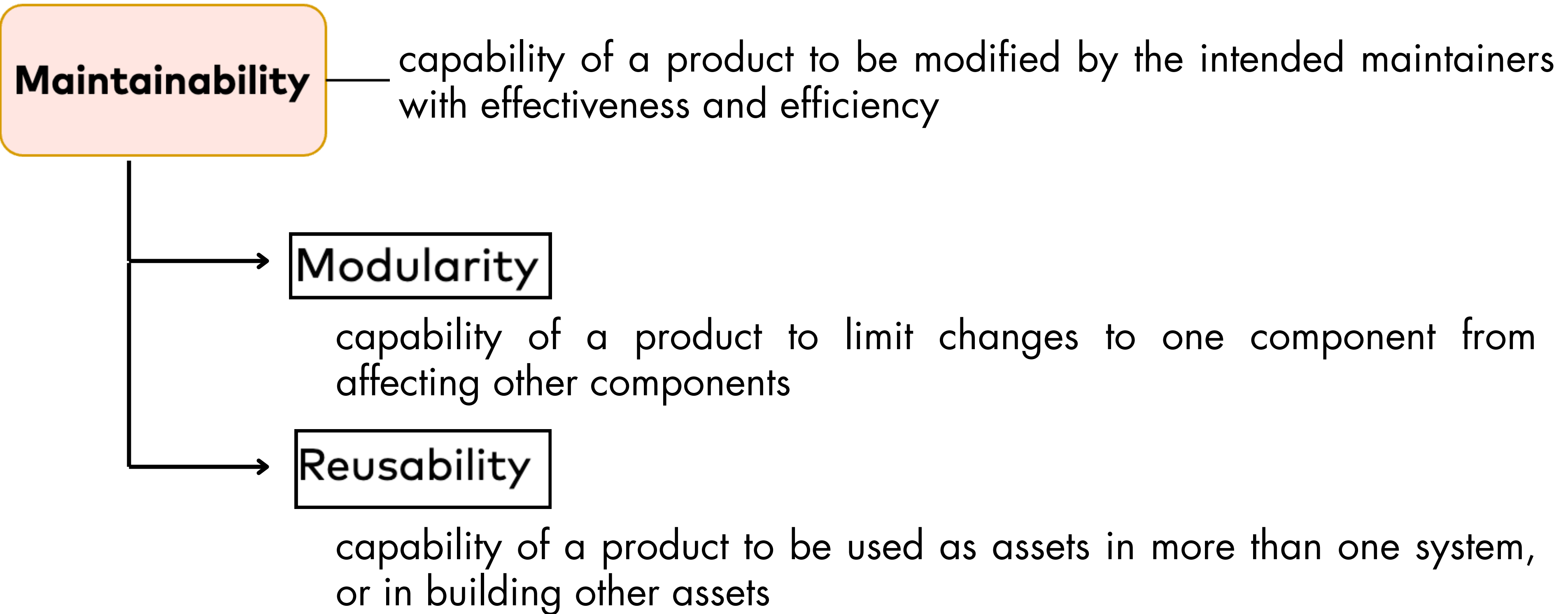
capability of a product to enable actions of an entity to be traced uniquely to the entity

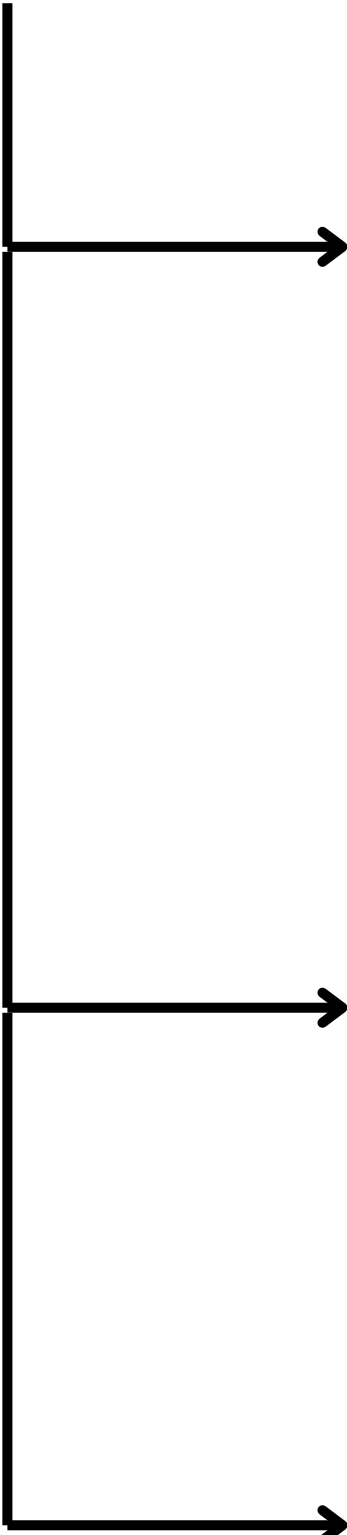
Authenticity

capability of a product to prove that the identity of a subject or resource is the one claimed

Resistance

capability of a product to sustain operations while under attack from a malicious actor





Analysability

capability of a product to be effectively and efficiently assessed regarding the impact of an intended change to one or more of its parts, to diagnose it for deficiencies or causes of failures, or to identify parts to be modified

Modifiability

capability of a product to be effectively and efficiently modified without introducing defects or degrading existing product quality

Testability

capability of a product to enable an objective and feasible test to be designed and performed to determine whether a requirement is met

Flexibility

capability of a product to be adapted to changes in its requirements, contexts of use, or system environment

Adaptability

capability of a product to be effectively and efficiently adapted for or transferred to different hardware, software or other operational or usage environments

Scalability

capability of a product to handle growing or shrinking workloads or to adapt its capacity to handle variability

Installability

capability of a product to be effectively and efficiently installed successfully and/or uninstalled in a specified environment

Replaceability

capability of a product to replace another specified product for the same purpose in the same environment

Safety

capability of a product under defined conditions to avoid a state in which human life, health, property, or the environment is endangered

Operational constraint

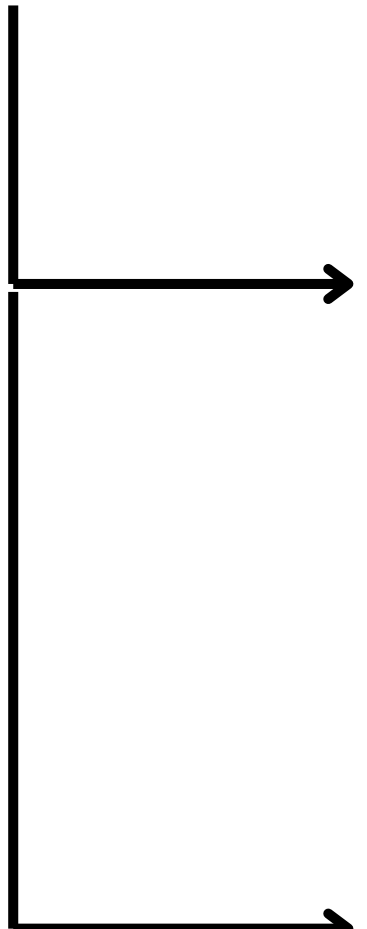
capability of a product to constrain its operation to within safe parameters or states when encountering operational hazard

- Risk identification

capability of a product to identify a course of events or operations that can expose life, property or environment to unacceptable risk

- Fail safe

capability of a product to automatically place itself in a safe operating mode, or to revert to a safe condition in the event of a failure



Hazard warning

capability of a product to provide warnings of unacceptable risks to operations or internal controls so that they can react in sufficient time to sustain safe operations

Safe integration

capability of a product to maintain safety during and after integration with one or more components